ETHNOHACK 2026 · TRACK 1 — TRADITIONAL KNOWLEDGE

Palimpsest

reading 2,000 years of herbal medicine against modern evidence

Dioscorides Greek, c. 60 CE **Culpeper** English, 1653 **Modern evidence** 2026

Bridges to Track 2 (drug discovery) and Track 3 (conservation)

Code & data github.com/KaraAliOsman/palimpsest

Explorer karaaliosman.github.io/palimpsest

Team Palimpsest · EthnoHack 2026

Open the explorer

Herbals are mined for leads, but resist being read

THE LANGUAGE

ἰκτερικούς θεραπεύει

"treats the jaundiced": inflected, accented Ancient Greek

THE PLANTS

ἀψίνθιον = ?

Artemisia absinthium? *A. pontica*? *A. maritima*?

THE DISEASES

dropsy · king's evil · strangury

today: oedema, scrofula, painful urination

THE EVIDENCE

"supported", compared with what?

earlier surveys counted supported claims with no base rate

Three questions

1. What persisted across 1,600 years of copying?
2. Does persistence carry a modern signal?
3. What has modern science ignored?

Three layers of text and evidence, 60 CE to 2026



Dioscorides, *De materia medica*

822

chapters

Wellmann's critical edition (First1KGreek TEI). We repaired a nesting bug that hid 199 chapters from parsers.

Culpeper, *The Complete Herbal*

328

monographs

Project Gutenberg #49513. Only the “government and virtues” sections are mined; astrology is ignored.

Modern literature and data

48.2 M

Europe PMC records

plus ClinicalTrials.gov, LOTUS phytochemistry via Wikidata, GBIF, IUCN status and MeSH

From the original words to one shared key

DIOSCORIDES · DE MATERIA MEDICA 3.23

ἀψίνθιον … καὶ ἀνορεξίας καὶ **ἰκτερικούς** θεραπεύει

wormwood … "and treats loss of appetite and the jaundiced"

CULPEPER · COMPLETE HERBAL · WORMWOOD

*"The sun never shone upon a better herb for the **yellow jaundice** than this"*

1653, "government and virtues"

1 NAME CONCORDANCE

ἀψίνθιον · Wormwood → ***Artemisia absinthium***

confidence high · all names: 99.2% on GBIF; 64% / 72% agree with independent Greek / English checks

2 INDICATION EXTRACTION

ικτερ- · yellow jaundice → Jaundice

accent-free, offset-preserving Greek stems · 17th-century English disease crosswalk

SHARED KEY

Artemisia absinthium × Jaundice

MeSH D007565

claimed in both → persistent

85

MeSH-linked indication concepts

10,399

claims, each with a verbatim span

BLIND REFERENCE EVALUATION · 50 + 49 EVALUABLE RANDOM MONOGRAPHS

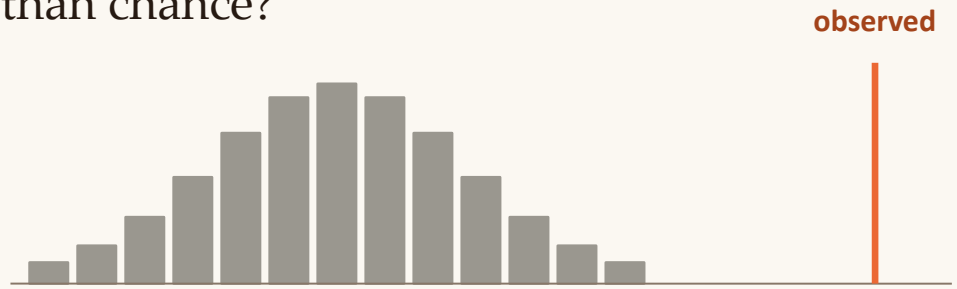
Greek precision 0.88 · recall 0.83 · F1 0.86 [0.81–0.89]
English precision 0.90 · recall 0.90 · F1 0.90 [0.88–0.92]

Reference annotation AI-assisted; human double annotation is next.

Each question is tested against its own control

Q1 · TRANSMISSION

Did Culpeper repeat Dioscorides more often than chance?



chance: 10,000 label shuffles

Culpeper's claim sets are shuffled among the shared plants: claims per plant and disease frequencies stay fixed.

Q2 · MODERN SIGNAL

Is a claimed use studied more than unclaimed uses of the same plant?

Artemisia absinthium



claimed — 12 random unclaimed indications, same plant

$$\log \mu = \log E + \beta \cdot group + \gamma_{indication}$$

Logit with indication fixed effects → odds of any study; Poisson GLM with volume offset → depth. Cluster-robust by plant.

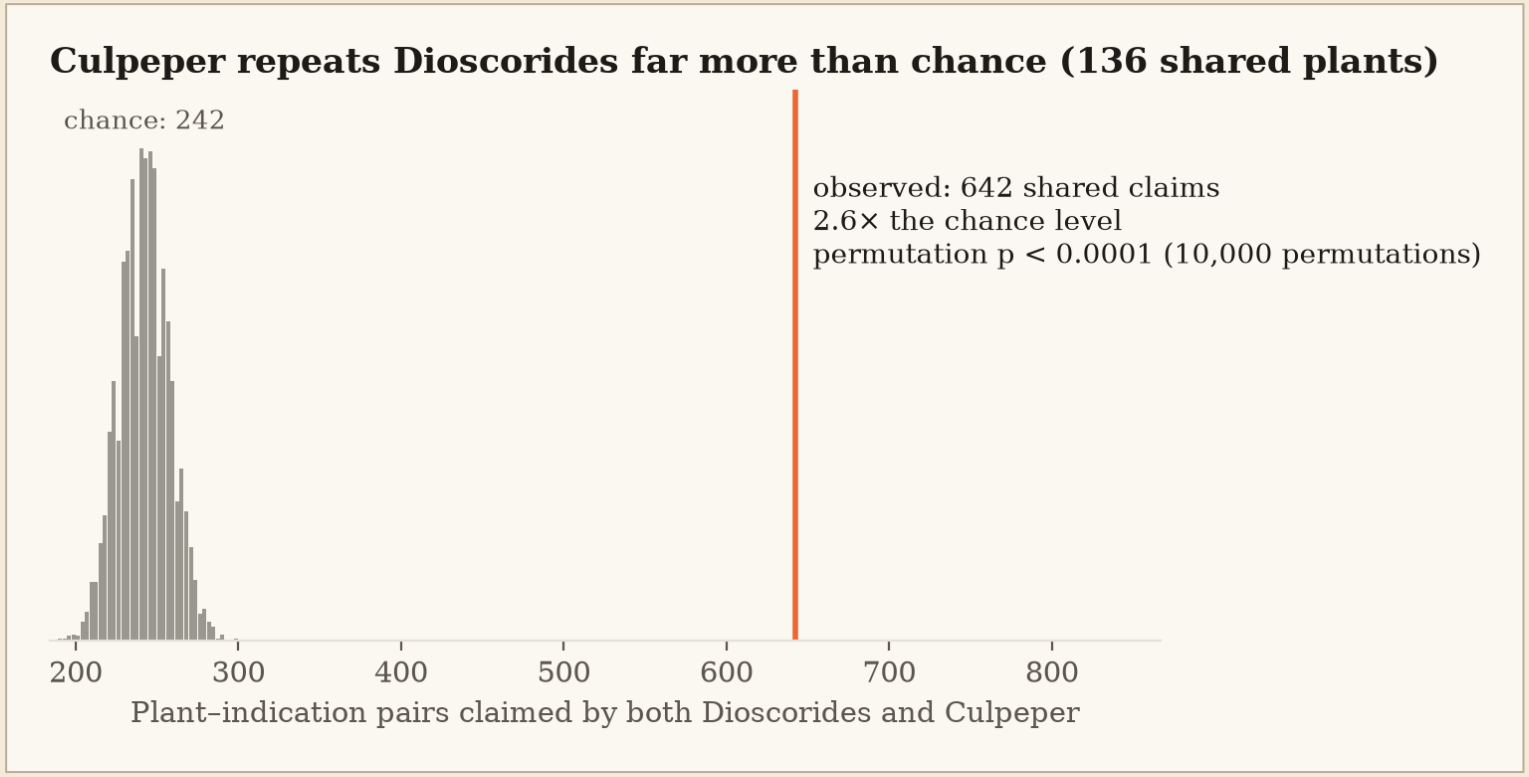
ROBUSTNESS

Genus level · high-confidence IDs only · therapeutic categories

TRANSPARENT LEAD SCORE

0.35 persistence + 0.25 chemistry + 0.25 gap + 0.15 tractability

Culpeper repeats Dioscorides 2.6× more often than chance



136 species in both herbals; 36% of Culpeper's claims for them were already in Dioscorides (13% expected by chance). Persistence of a textual tradition, not independent rediscovery.

642

plant–indication pairs claimed by both authors

242

expected if Culpeper's claims were shuffled among the same plants

$p < 0.0001 \cdot z = 26.1$

one-sided, 10,000 permutations

2.1× even when matching plants by entry length; 1.6× even within the same botanical family (both $p < 0.001$)

Genus level 2.39× · high-confidence IDs 2.49×

Claimed uses are more often studied, not studied more

AT LEAST ONE CO-MENTIONING PAPER

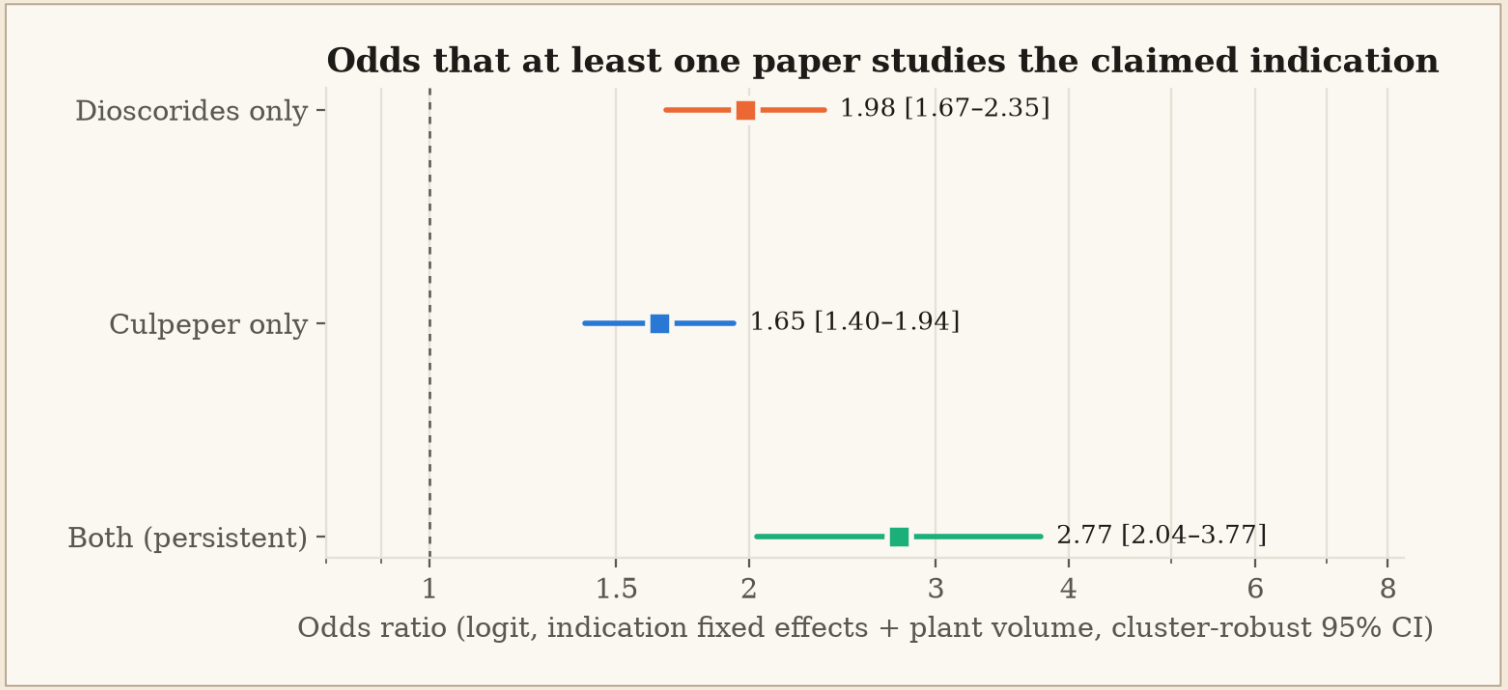
45% vs 32%

persistent claims vs matched unclaimed indications of the same plants

Both herbals	45% vs 32% · OR 2.77
Dioscorides only	37% vs 25% · OR 1.98
Culpeper only	35% vs 24% · OR 1.65

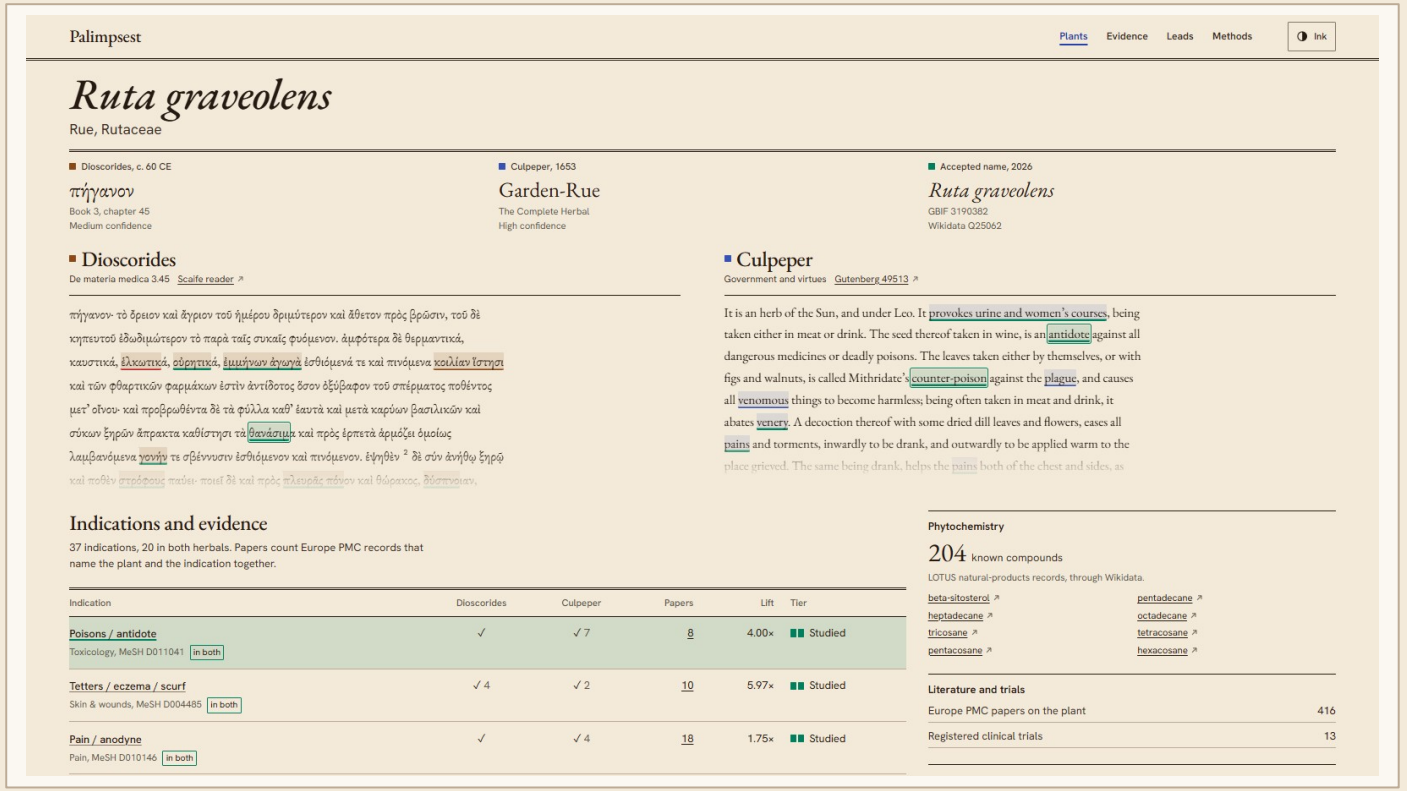
BUT NOT IN DEPTH

Once studied, not studied more: rate ratio 1.04 [0.61–1.76], n.s. That gap is what the leads target.

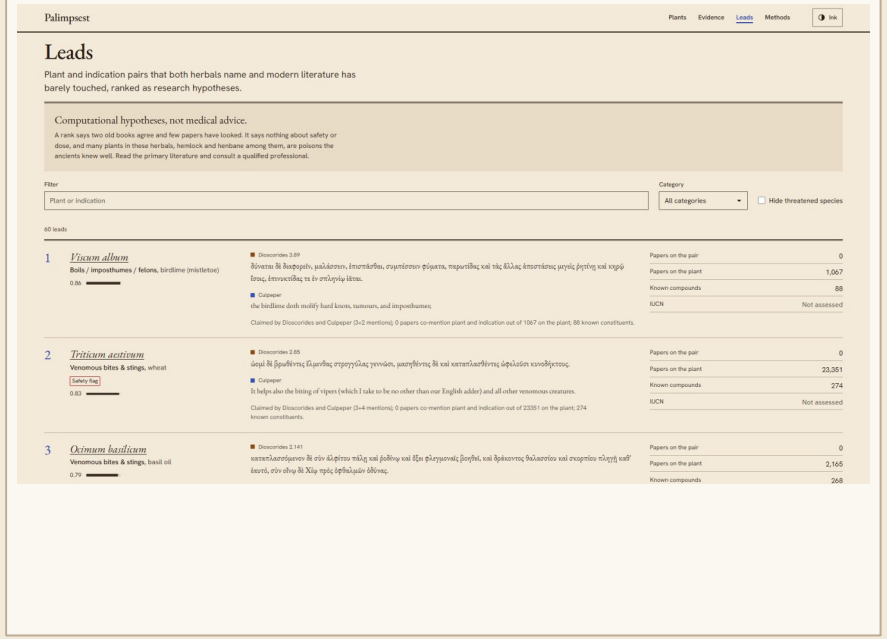


Logit with indication fixed effects and plant research volume, cluster-robust 95% CI; within-plant Wilcoxon $p = 10^{-15}$. Co-mention measures research attention, not efficacy.

Every number links back to the passage it came from



Plant page Dioscorides' Greek with each claim highlighted in place, Culpeper alongside, modern evidence per indication



Leads ranked, filterable, each score broken into its parts

60 persistent but under-studied leads

#	Plant	Indication	D + C	Papers	Compounds	Safety
1	*Ruta graveolens*	Eye complaints / dim sight	2 + 1	0	204	flag
2	*Vicia faba*	Bleeding / staunching blood	1 + 1	0	297	flag
3	*Ocimum basilicum*	Venomous bites & stings	2 + 3	0	268	–
4	*Juglans regia*	Ulcers & sores	1 + 4	0	190	flag
5	*Cannabis sativa*	Ear-ache / running ears	1 + 2	1	528	flag

Top 5 of 60 diversified leads from 517 candidates: persistent pairs with ≤ 4 co-mentioning papers. D + C = mentions in Dioscorides + Culpeper; compounds from LOTUS. Flag = the historical text itself warns of harm (shown, never scored).

Hypotheses for laboratory follow-up, not therapeutic claims.

Drug discovery · Track 2

An explainable shortlist with LOTUS constituents attached

Conservation · Track 3

No lead depends on a species assessed as threatened (19 of 60 assessed by IUCN)

ETHCSTWIN knowledge graph

20,425 nodes, 51,430 edges: 10,399 claims with quote and offsets, 18,498 compounds

History of medicine

Transmission measured passage by passage, from the Greek

What these results are not, and what comes next

Responsible research

- Computational hypotheses, never medical advice; several remedies are toxic
- Only published, openly licensed texts; nothing collected from living communities
- AI-assisted curation disclosed; every row has a confidence level and a justification

Limitations

- Circularity: tradition motivates research, so co-mention ≠ efficacy
- A co-mention may report a null or toxic result
- Ancient plants are identified retrospectively
- Culpeper inherits Dioscorides: persistence, not rediscovery
- Literature searched by accepted binomial only (synonyms implemented, not yet run: rosemary 211 → 2,179 papers)

Next

- Add Galen, Pliny, Byzantine and Arabic sources to trace the whole chain
- Outcome-aware evidence: classify findings, add ChEMBL targets
- Expert review with ETHCSTWIN, then lab follow-up of top leads

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Code MIT · Greek-derived data CC BY-SA 4.0